

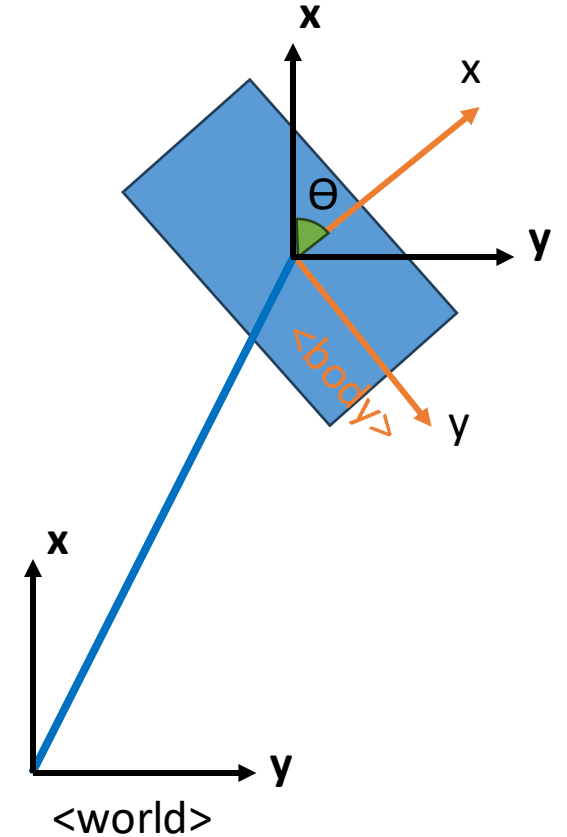
CLMVS – Solution of lab2

Controllability

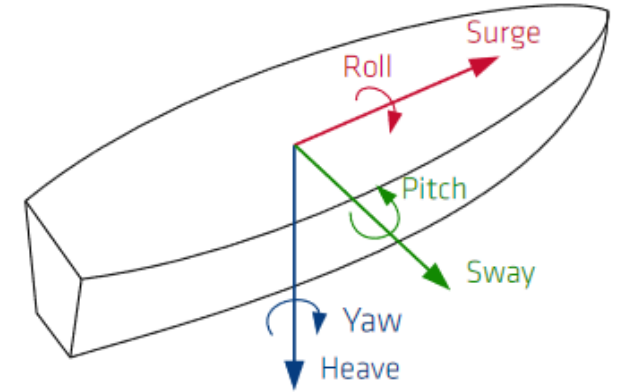
$$\begin{aligned} m\ddot{x} &= -\alpha\dot{x} + F_x \\ m\ddot{y} &= -\beta\dot{y} + F_y \\ J\dot{w} &= -\gamma w + N_z \end{aligned}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\frac{d}{dt}x = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{bmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ w \end{bmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix}$$

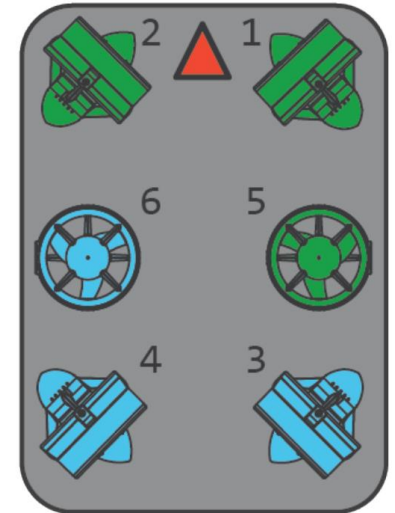


Thruster allocation matrix



Thruster allocation matrix (TAM)

	T1	T2	T3	T4	T5	T6
Surge	0.7071	0.7071	-0.7071	-0.7071	0	0
Sway	-0.7071	0.7071	-0.7071	0.7071	0	0
Heave	0	0	0	0	-1	1
Roll	0	0	0	0	0.218	0.218
Pitch	0	0	0	0	0.12	-0.12
Yaw	-0.1888	0.1888	0.1888	-0.1888	0	0

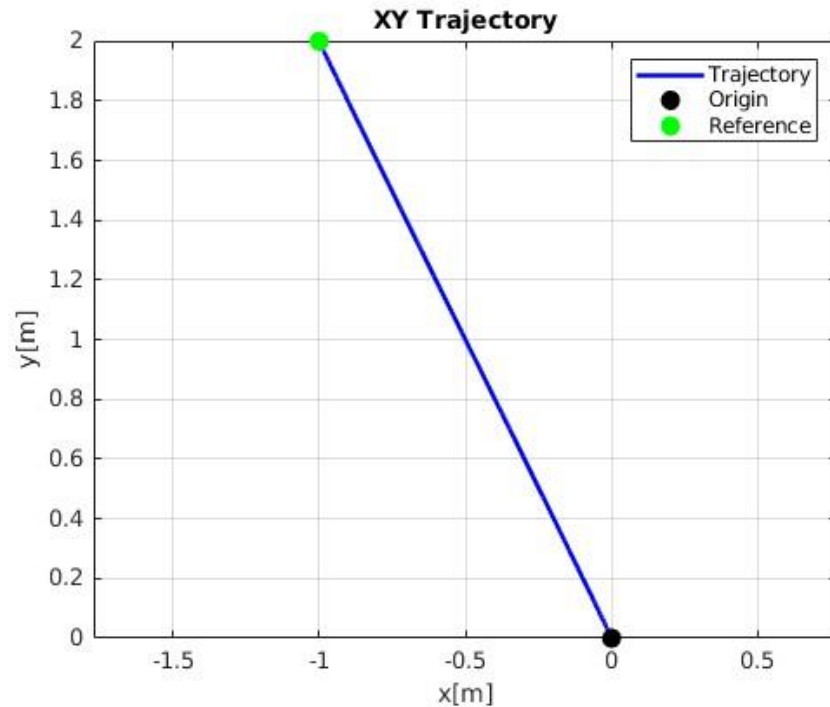


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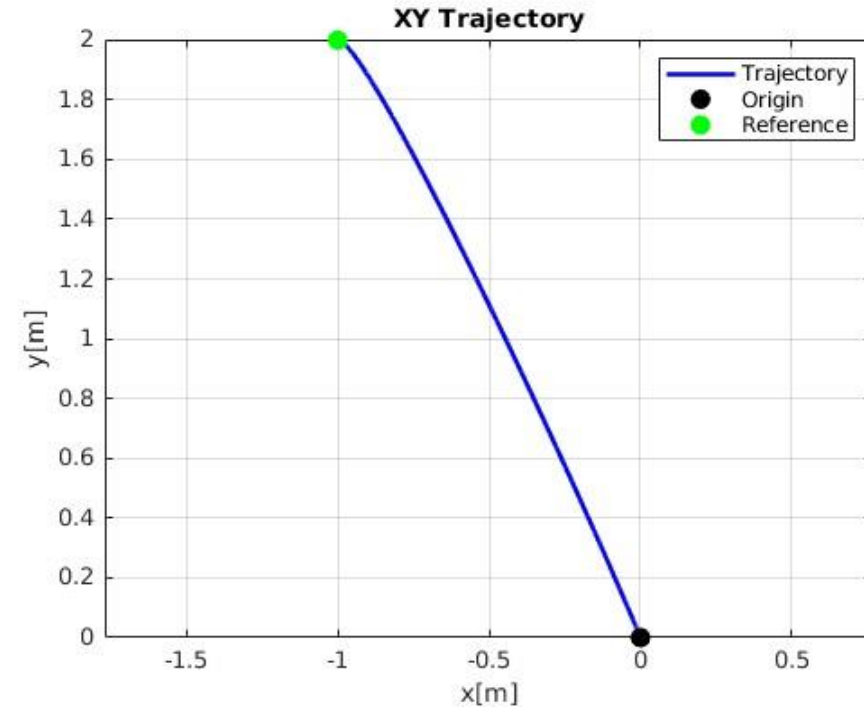
Controllability

3 cases:

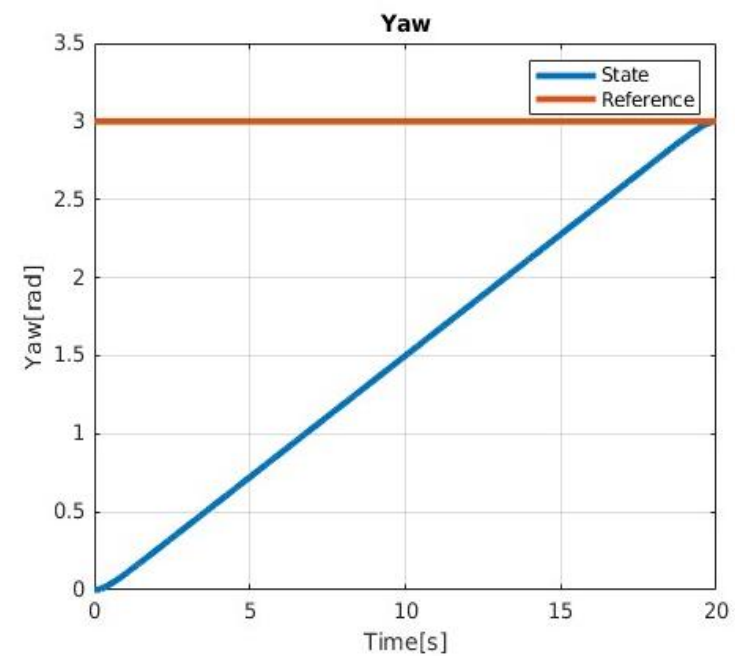
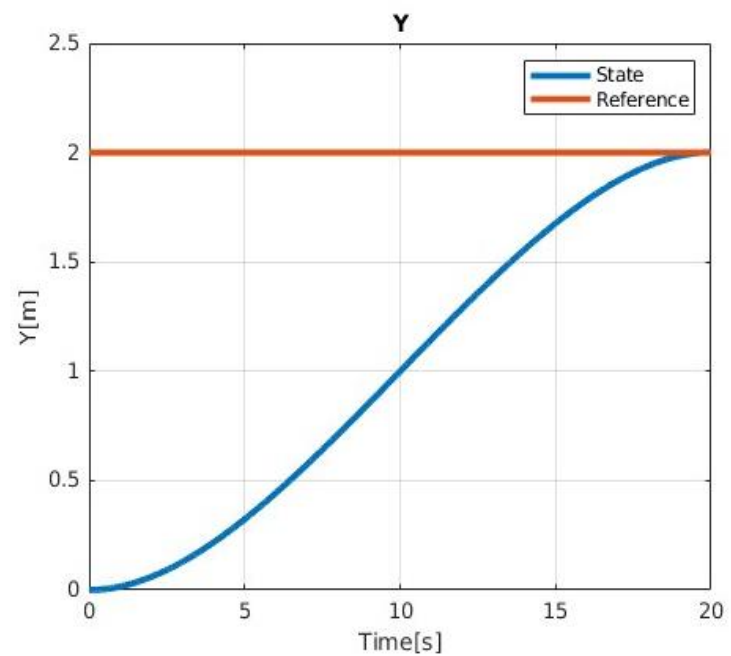
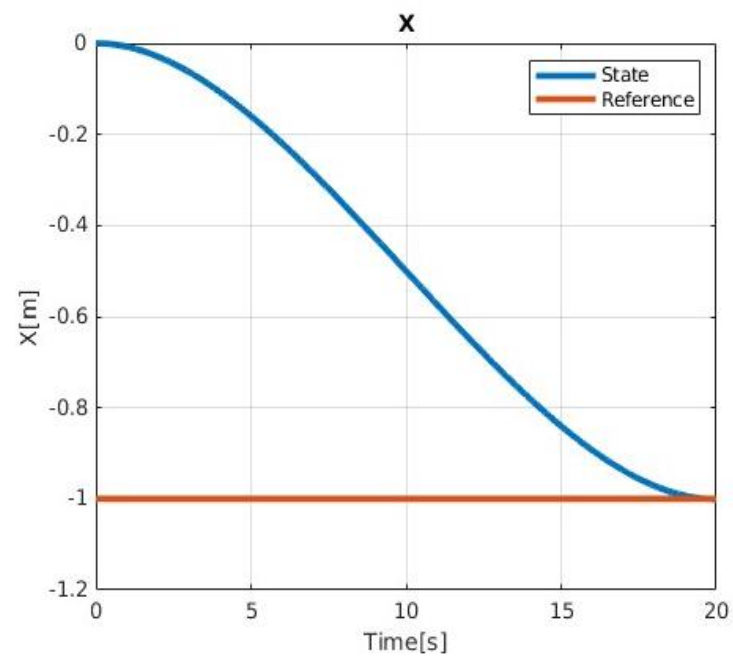
1. All thrusters enabled: fully controllable
2. Thruster 2 disabled: fully controllable (thanks to redundancy)



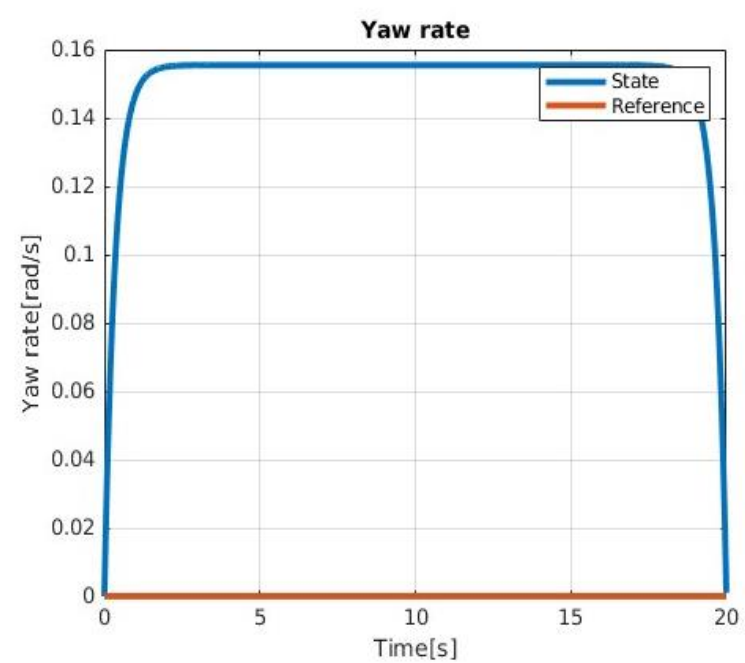
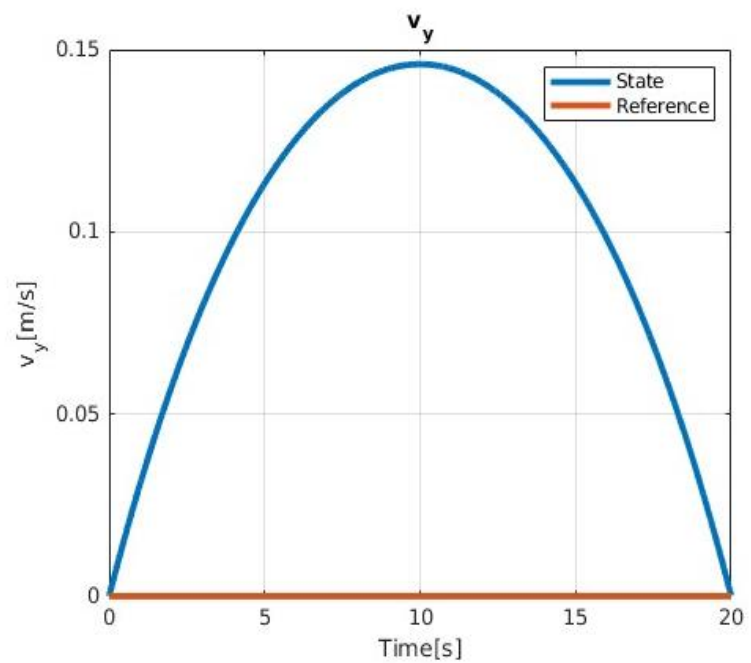
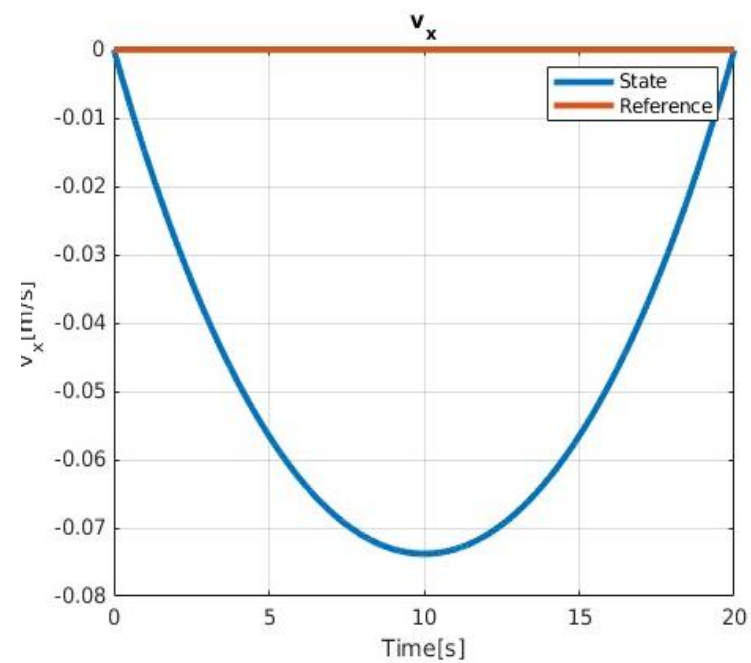
Case 1

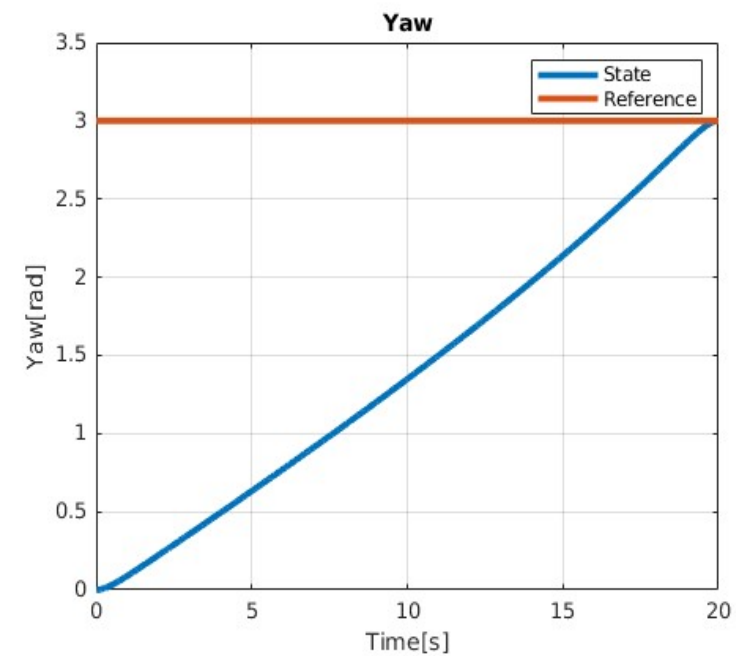
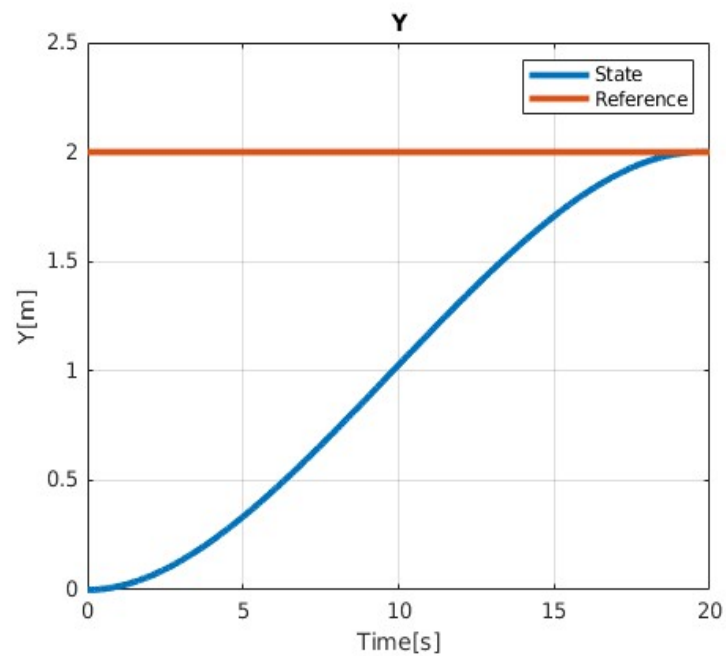
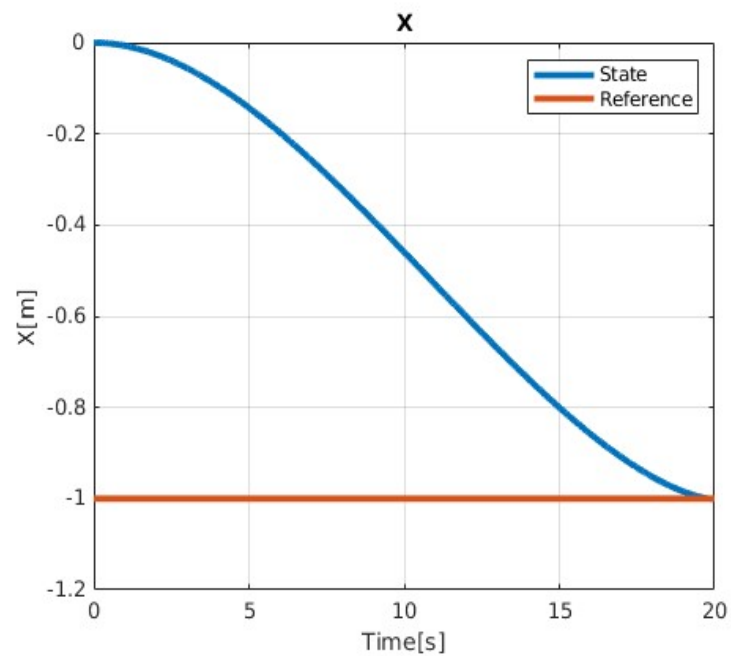


Case 2

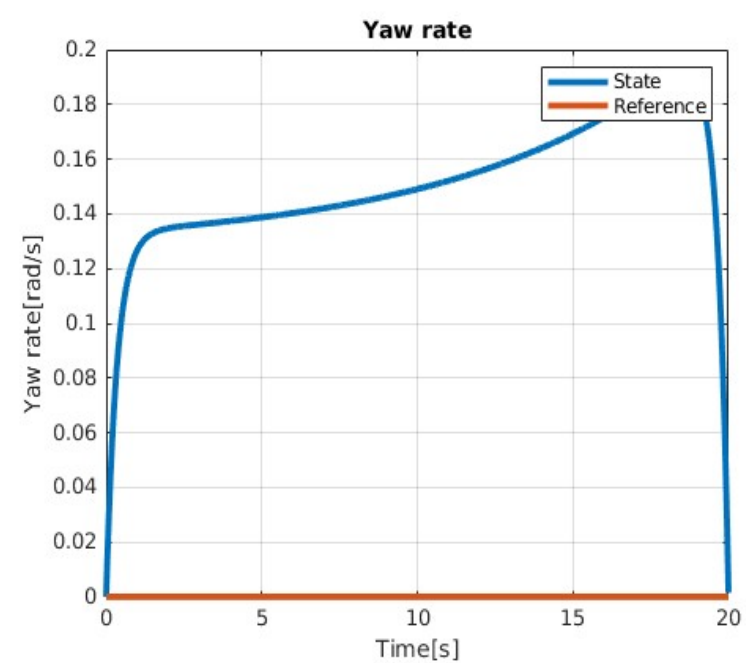
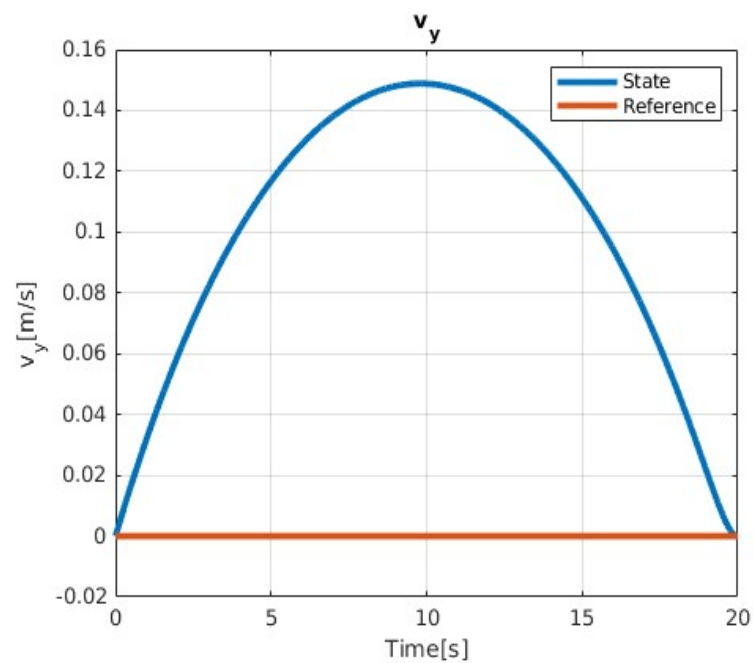
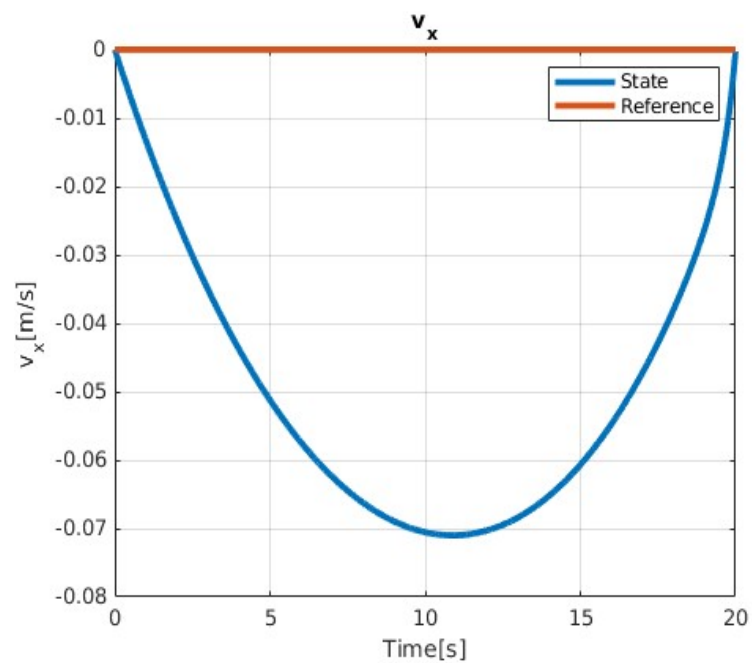


Case 1





Case 2



CLMVS – Solution of lab2

Controllability

3. Thrusters 2,3 disabled: not fully controllable ($\text{rank}(\mathbf{G}) = 5$)
What is controllable in this case?

Take the SVD of the controllability Gramian $\mathbf{G}$:

$$[\mathbf{U}, \mathbf{D}, \mathbf{V}] = \text{svd}(\mathbf{G})$$

The first $\text{rank}(\mathbf{G})$ columns of $\mathbf{U}$ are an orthonormal basis of the controllable subspace. Let's analyze $\mathbf{U}$...

$\mathbf{D}$ contiene i valori singolari (energia nelle direzioni dello stato)

$\mathbf{U}$ contiene le direzioni nello spazio degli stati

$\mathbf{V}$ contiene le direzioni nello spazio degli ingressi

Se un valore singolare è:

grande $\rightarrow$ direzione dello stato facilmente controllabile

piccolo $\rightarrow$ direzione difficile da controllare

zero $\rightarrow$ direzione non controllabile (ULTIMO VALORE DI $\mathbf{D}$)

CLMVS – Solution of lab2

Controllability

colonne 2 e 3 di $U \rightarrow$ solo θ e $\omega \rightarrow$ sottospazio di rotazione completamente controllabile

colonne 1,4,5 $\rightarrow$ controllano combinazioni di (x, y, v_x, v_y) , ma non tutte indipendenti

U =		Dir. 1			controllable subspace		Dir. NON CONT	
x	-0.7551	-0.0000	0.0000	-0.1656	0.6302	0.0728		
y	0.6539	-0.0000	0.0000	-0.1214	0.7409	0.0936		
z	0	0.9996	0.0277	0	0	0		
v_x	-0.0373	0.0000	0.0000	0.7085	0.0603	0.7021		
v_y	0.0284	0.0000	0.0000	-0.6752	-0.2244	0.7021		
v_z	-0.0000	0.0277	-0.9996	0.0000	0.0000	0.0000		

- Columns 2 and 3 only have non-zero components on the third and sixth row (θ, ω)
- Moreover, they are not proportional and therefore are linearly independent
- This means that whole rotation subspace is controllable, that is you can rotate by any specified amount and with any target final angular velocity

CLMVS – Solution of lab2

Controllability

U =

controllable subspace					
-0.7551	-0.0000	0.0000	-0.1656	0.6302	0.0728
0.6539	-0.0000	0.0000	-0.1214	0.7409	0.0936
0	0.9996	0.0277	0	0	0
-0.0373	0.0000	0.0000	0.7085	0.0603	0.7021
0.0284	0.0000	0.0000	-0.6752	-0.2244	0.7021
-0.0000	0.0277	-0.9996	0.0000	0.0000	0.0000

- Columns 1, 4 and 5 are linearly independent (check!)
- However, the subspace $(x, y, \dot{x}, \dot{y})$ is 4-dimensional
- 3 vectors alone are not sufficient to span a 4-dimensional vector space
- This means that you can reach any final position, but not with arbitrary final velocity (or viceversa)

CLMVS – Solution of lab2

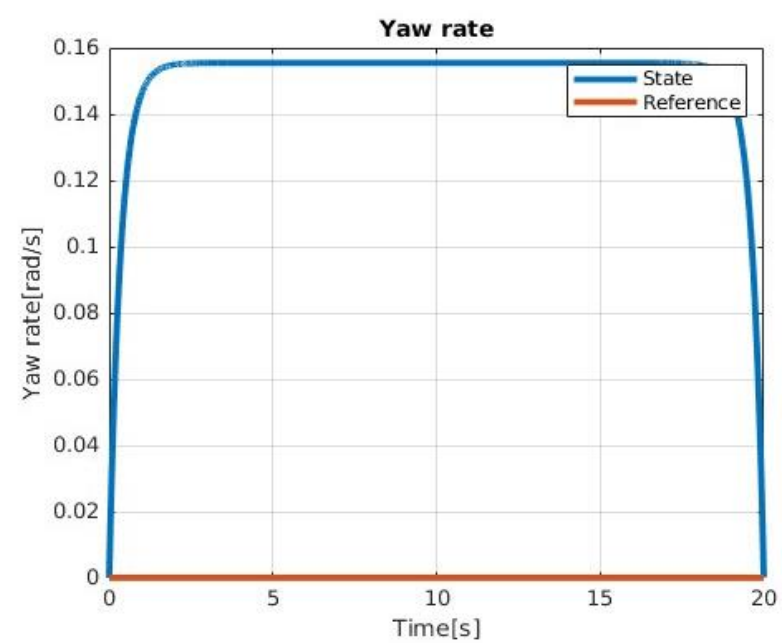
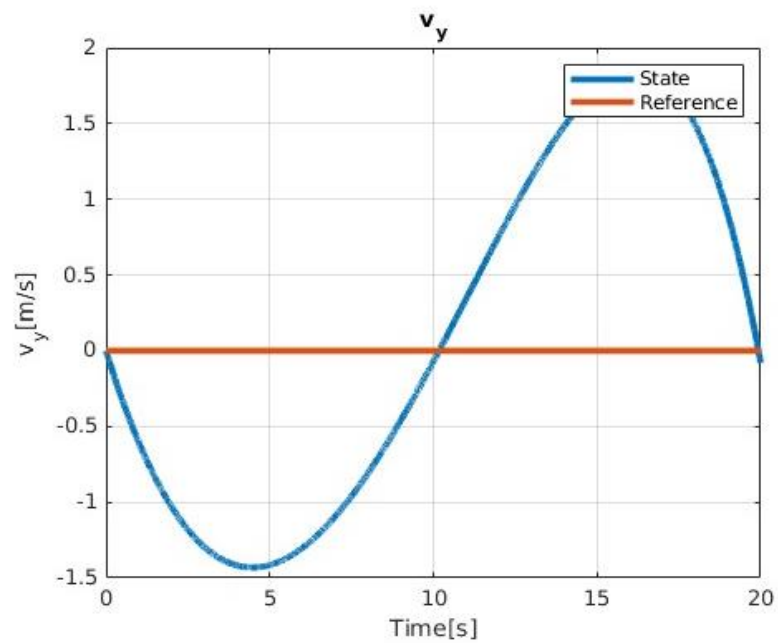
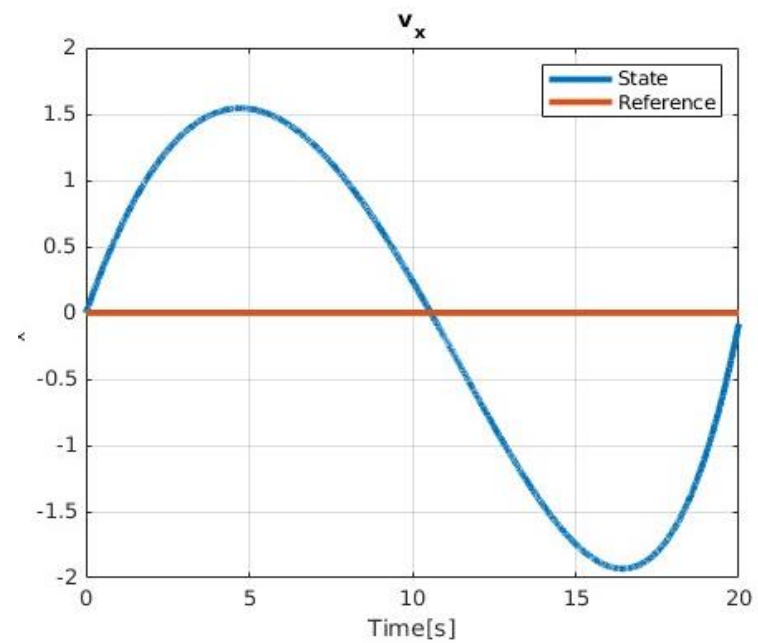
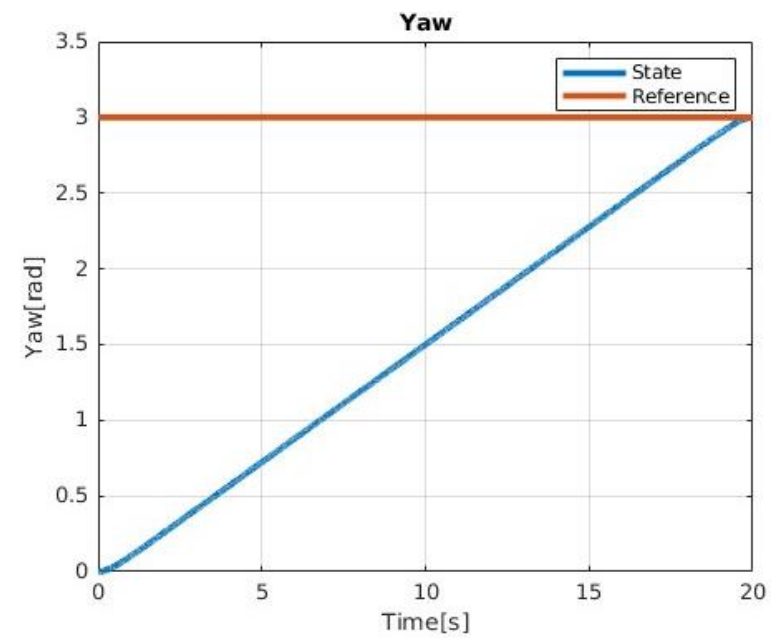
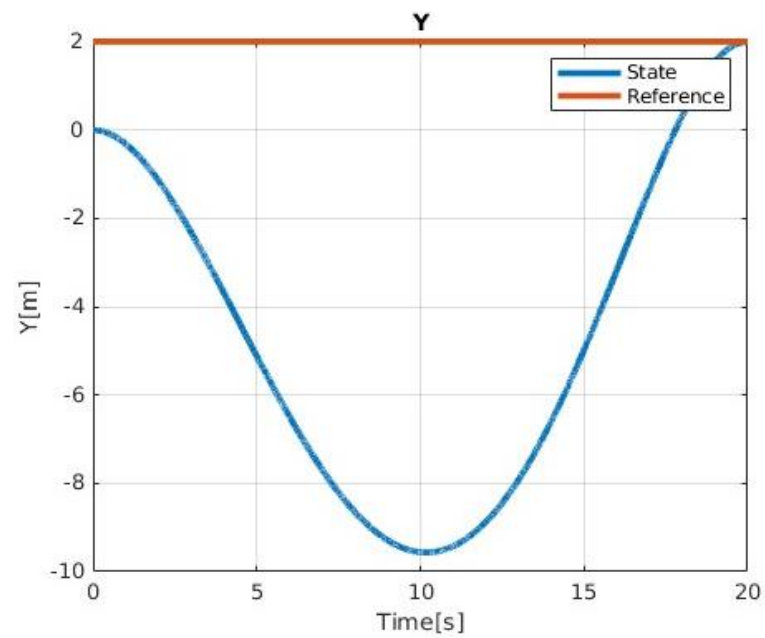
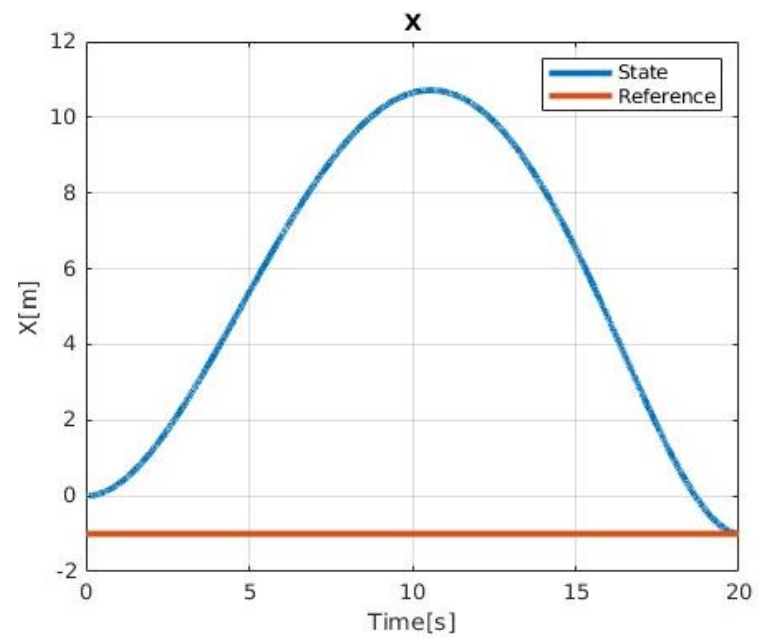
Controllability

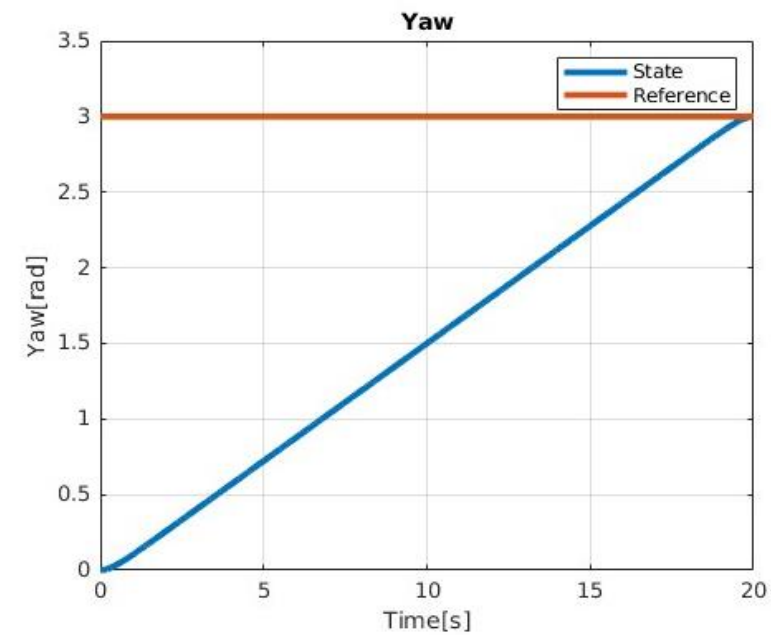
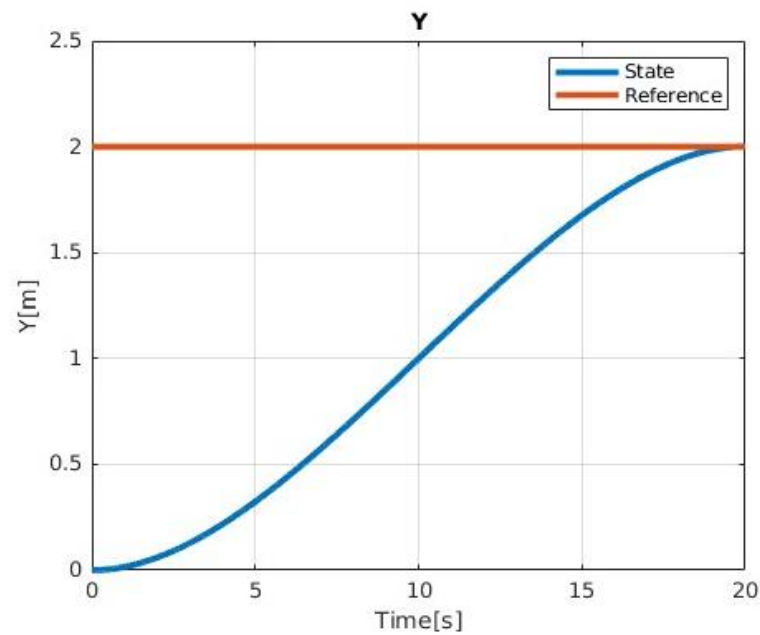
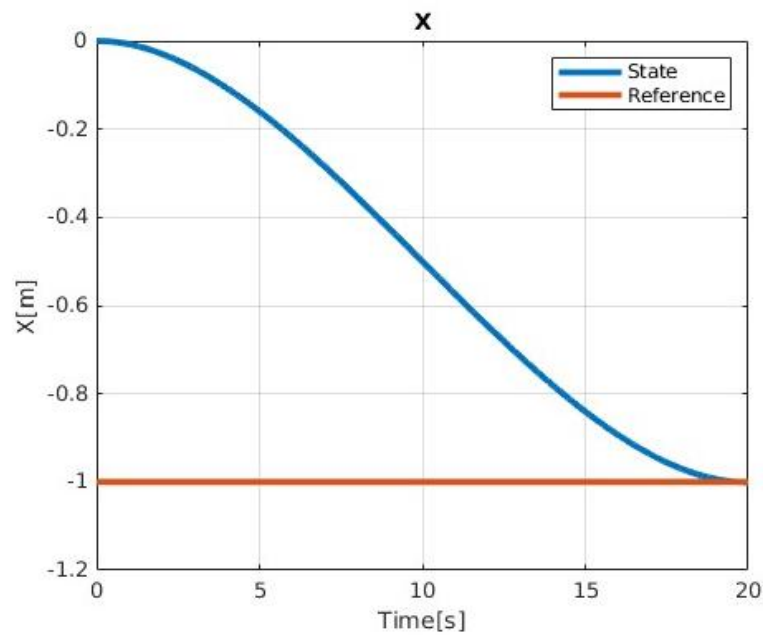
- What happens if we still try to compute the system input from the controllability Gramian in the not fully controllable case (replacing $\text{inv}(G)$ with $\text{pinv}(G)$)?
- Expectations?

Le prime 5 colonne di U rappresentano le 5 direzioni dello stato che il sistema può controllare anche con due thruster disattivati.

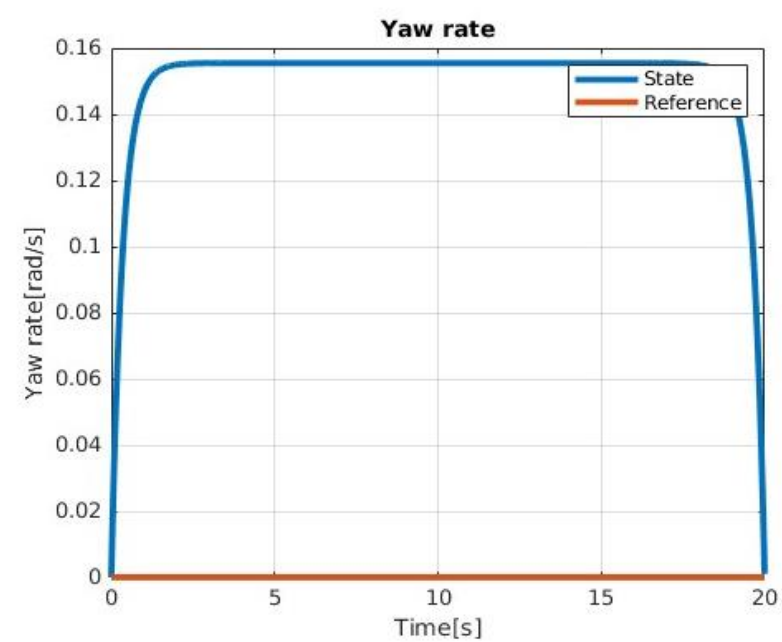
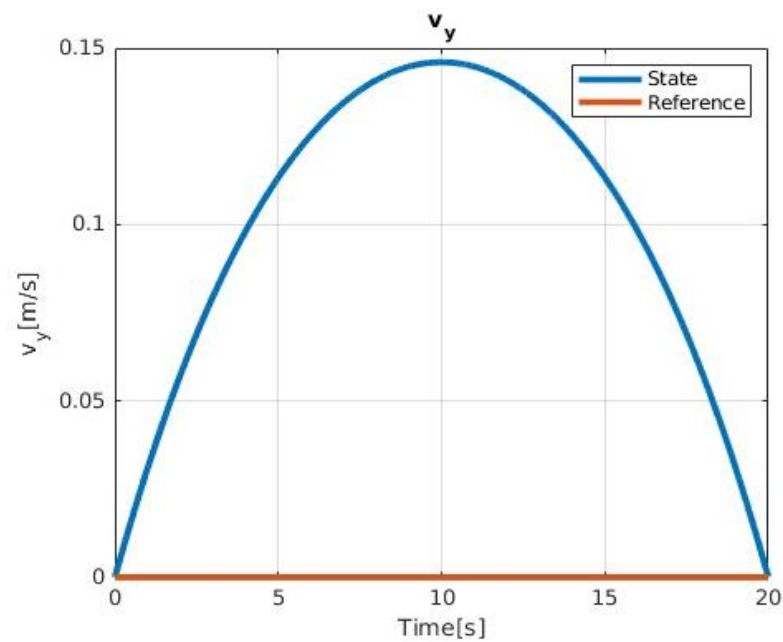
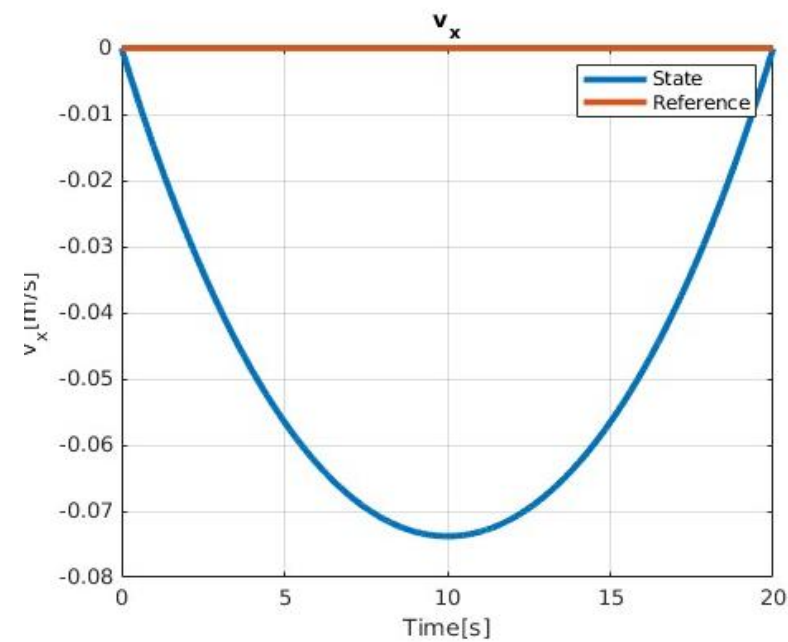
La parte rotazionale (θ, ω) è chiaramente dentro lo spazio controllabile, mentre le posizioni e le velocità lineari sono controllabili solo in combinazioni specifiche.

L'ultima colonna di U , associata al valore singolare nullo, rappresenta la direzione non controllabile: il sistema non può imporre arbitrariamente quella combinazione di stati.





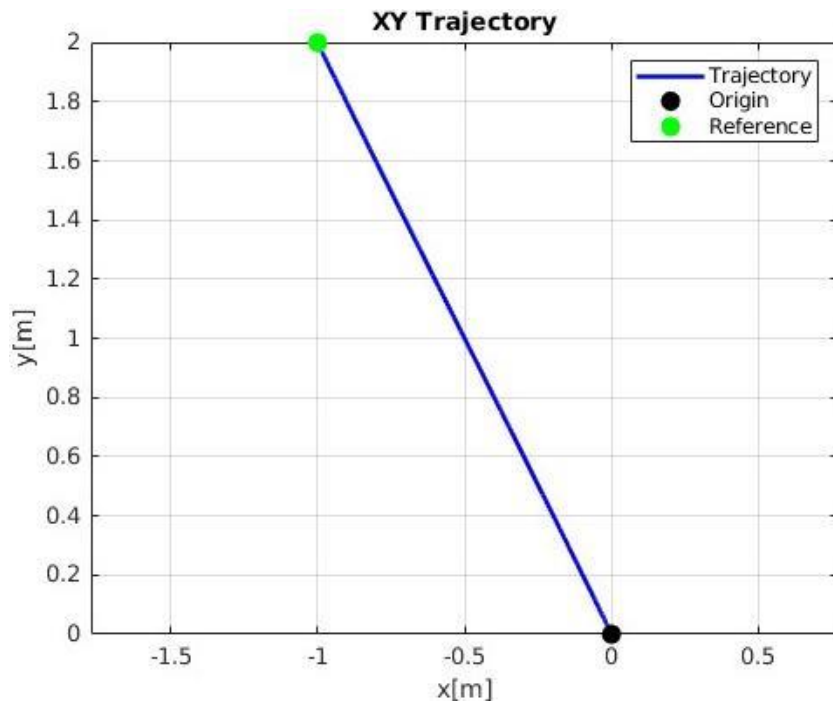
Compare to the controllable case



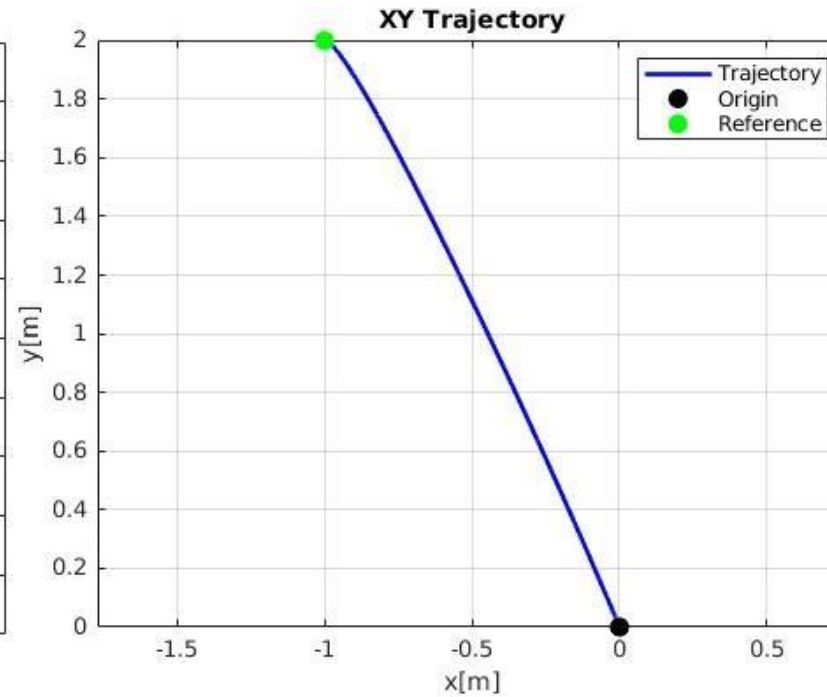
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Controllability

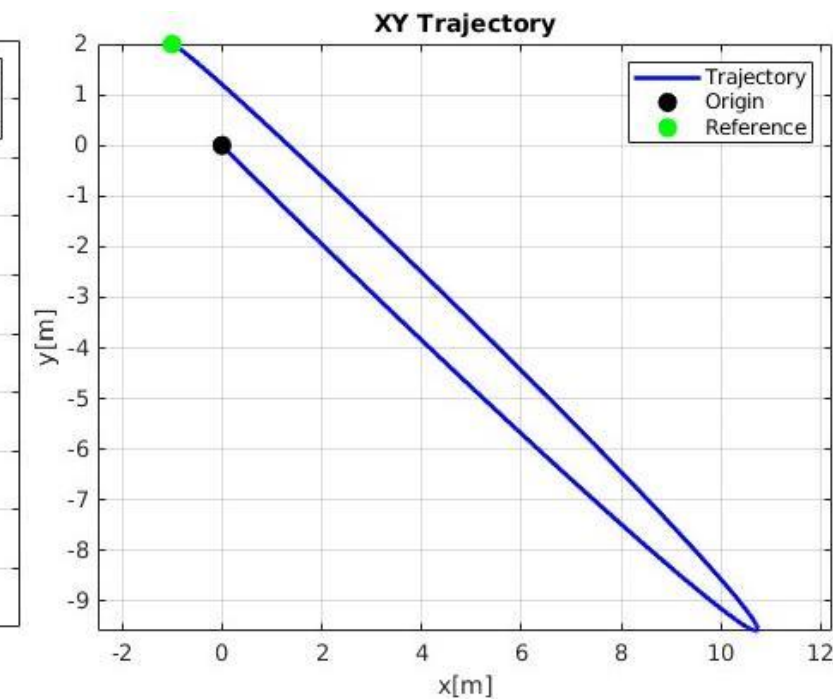
- Trajectory comparison:



Case 1



Case 2



Case 3

CLMVS – Solution of lab2

Controllability

- Numerically the final error (using $dt = 1e-3$) is:

$1.0e-03$ *

-0.0881
0.1585
0.1165
-0.0069
0.0123
0.1103

Case 1

$1.0e-03$ *

-0.0811
0.1614
0.1172
-0.0253
-0.0053
0.1480

Case 2

-0.0056
-0.0127
0.0001
-0.0809
-0.0798
0.0001

Case 3

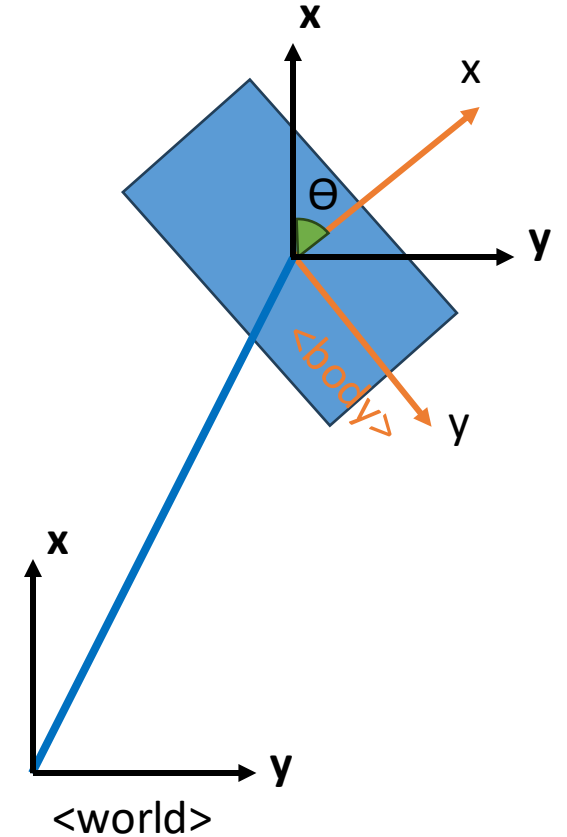
CLMVS – lab3

Observability

$$\begin{aligned} m\ddot{x} &= -\alpha\dot{x} + F_x \\ m\ddot{y} &= -\beta\dot{y} + F_y \\ J\dot{w} &= -\gamma w + N_z \end{aligned}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\frac{d}{dt}x = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{bmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ w \end{bmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix}$$



Computing Ko

Assume the positions are measured.

Since the observer is the positions and the orientation, which are coupled with their derivatives (the velocities), again we can deal with each two equation (position, velocity) as one independent system.

New state definition:

$$x \triangleq \begin{pmatrix} r_x \\ v_x \end{pmatrix} = \begin{pmatrix} x_1 \\ x_4 \end{pmatrix}$$

$$\begin{cases} \dot{r}_x = v_x \\ \dot{v}_x = -\frac{\alpha}{m} v_x + \frac{F_x}{m} \end{cases}$$

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha/m \end{bmatrix} \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F_x$$

$$y = [1 \quad 0] \begin{pmatrix} x_1 \\ x_4 \end{pmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha/m \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, C = [1 \quad 0]$$

Computing K_o

DIMOSTRAZIONE LUENBERGER ESTIMATOR

$$\dot{\hat{x}} = A\hat{x} + Bu_1 + K_{o_{surge}}(y - \hat{y}); K_{o_{surge}} = \begin{bmatrix} k_{x_1} \\ k_{x_4} \end{bmatrix}$$

$$\dot{x} = Ax + Bu_1$$

$$\tilde{x} = x - \hat{x}; \text{ errore di stima}$$

$$\dot{\tilde{x}} = \dot{x} - \dot{\hat{x}} = Ax + Bu_1 - A\hat{x} - Bu_1 - K_{o_{surge}}(y - \hat{y})$$

$$\dot{\tilde{x}} = A\tilde{x} - K_{o_{surge}}C\tilde{x} = (A - K_{o_{surge}}C)\tilde{x}$$

We should choose K such that $(A - K_{o_{surge}}C)$ is asymptotically stable. $(A-KC) < 0$

$$A - K_{o_{surge}}C = \begin{pmatrix} 0 & 1 \\ 0 & -\frac{\alpha}{m} \end{pmatrix} - \begin{pmatrix} k_{x_1} & 0 \\ k_{x_4} & 0 \end{pmatrix} = \begin{pmatrix} -k_{x_1} & 1 \\ -k_{x_4} & -\frac{\alpha}{m} \end{pmatrix}$$

Computing K_o

$$\det(\lambda I - [A - K_{o_{surge}} C]) = \begin{vmatrix} \lambda + k_{x_1} & -1 \\ k_{x_4} & \lambda + \frac{\alpha}{m} \end{vmatrix}$$

$$= \left(\lambda + \frac{\alpha}{m}\right)(\lambda + k_{x_1}) + k_{x_4}$$

$K_{o_{surge}}$

$$= \lambda^2 + \left(\frac{\alpha}{m} + k_{x_1}\right)\lambda + \left(k_{x_1}\frac{\alpha}{m} + k_{x_4}\right) = 0$$

$$\begin{cases} k_{x_1}\frac{\alpha}{m} + k_{x_4} = \omega_n^2 \\ \frac{\alpha}{m} + k_{x_1} = 2\zeta\omega_n \end{cases}$$

TROVO I DUE AUTOVALORI
DI A-KC (che corrispondono ai
poli da scegliere per progettare
correttamente K)

For our exercise we can choose $\zeta = 1$, $\omega_n = 1$ rad/s

Computing K_o

- We can repeat the same method for $[r_y, v_y]$, $[\theta, w]$ to compute all components of K_o

$$\bullet K_o = \begin{bmatrix} k_{x_1} & 0 & 0 \\ 0 & k_{x_2} & 0 \\ 0 & 0 & k_{x_3} \\ k_{x_4} & 0 & 0 \\ 0 & k_{x_5} & 0 \\ 0 & 0 & k_{x_6} \end{bmatrix}$$

$$\text{surge: } \begin{cases} k_{x_1} \frac{\alpha}{m} + k_{x_4} = w_n^2 \\ \frac{\alpha}{m} + k_{x_1} = 2\zeta w_n \end{cases}$$

$$\text{sway: } \begin{cases} k_{x_2} \frac{\beta}{m} + k_{x_5} = w_n^2 \\ \frac{\beta}{m} + k_{x_2} = 2\zeta w_n \end{cases}$$

$$\text{yaw: } \begin{cases} k_{x_3} \frac{\gamma}{J_w} + k_{x_6} = w_n^2 \\ \frac{\gamma}{J_w} + k_{x_3} = 2\zeta w_n \end{cases}$$

Exercise

You are given the template script *lab3.m*.

1. Verify observability in case position variables only are measured.
2. Implement a Luenberger observer, assuming that the state is observable, being the position variables only measured, tuning the observer gain matrix K_o . Assume that the initial state is known with error $[1, -1, 1.5, 0.5, -0.5, -1.5]$. Plot the difference between the true and estimated state.
3. Repeat 1,2 assuming that the whole state is measured.

se misuri tutto lo stato, l'osservatore diventa concettualmente inutile: puoi ricostruire X direttamente da Y

